

Module: Corporate Finance & Equity Valuation

School of Economics

University of Cape Town

Capital Structure

Learning Objectives

1. Describe the two Miller & Modigliani propositions, the key assumptions underlying them and their relevance to capital structure decisions.
2. Discuss the benefits and costs of using debt financing.
3. Describe the trade-off and pecking order theories of capital structure choice and explain what the empirical evidence tells us about these theories.
4. Discuss some of the practical considerations that managers are concerned with when they choose a firm's capital structure.

Capital Structure

- Capital structure is the mix of debt and equity that is used by the company to finance its operations.
- The mix will always include ordinary shares and may often include debt and preference shares.
- The firm may have several classes of ordinary shares (class A), for example with different voting rights and possibly different claims on the cash flows available to shareholders.

Capital Structure

- Equity refers to the funds contributed by shareholders.
- Equity holders/shareholders earn dividends.
- The cost of equity is the required rate of return by shareholders.

Capital Structure

- Preference shares can be cumulative or non-cumulative and convertible or none convertible into ordinary shares.
- Debt includes all borrowed funds.
- Debt can be either long or short term (shorter term debt is riskier but allows greater flexibility).
- Debt can be either secured using other assets of the company, or unsecured.
- Secured debt is riskier as the borrowing company has less flexibility with regards to the assets.

Capital Structure

- The use of debt exposes a company to both interest and capital risk.
- The fraction of the total financing that is represented by debt is a measure of the financial leverage in the firm's capital structure.
- A higher fraction of debt indicates a higher degree of financial leverage.
- The amount of financial leverage in a firm's capital structure is important because it affects the value of the firm.
- Debt increases fixed costs and can lead to either positive or negative leverage.
- Leverage increases the volatility of earnings.

The Optimal Capital Structure

- What is the primary goal of financial management?
 - ❖ Maximizing shareholders' wealth.
- Meaning: The Financial manager should choose a Capital Structure that maximizes shareholders' wealth.
- Shareholders 'wealth can be maximized by maximizing firm value or minimizing Weighted average cost of capital (WACC).
- Remember: WACC is the discount rate , thus, the smaller or lower the discount rate, the greater the present value of the Free Cash Flows to the Firm (FCFTF).
- Note that: Firm value is equal to the PV of future Cash Flows to the firm discounted at WACC
- Since firm value and discount rate move in opposite directions, firm value is maximized when WACC is minimized.

The Optimal Capital Structure

- Determining an 'optimal' capital structure – i.e. balance between debt and equity financing – relies on the basic principle of risk and return.
- Use of debt in the company's capital structure exposes the company to financial risk (interest risk and capital risk).
- Question: Is the use of debt a good or bad thing?
- The good
 - ❖ Lower cost (cost of debt is lower)
 - ❖ Tax deductible
 - ❖ Easier to obtain

The bad

Cash outflow

Compulsory payments

Shareholders require higher returns due to increased risk.

The Optimal Capital Structure

- Determining an 'optimal' capital structure – i.e. balance between debt and equity financing – relies on the basic principle of risk and return.
- Increasing debt finance increases financial leverage, therefore driving up the required rate of return demands by incremental debt and equity providers.
- However, introducing debt – especially at low levels of debt – will generate incremental returns that may well outweigh any increased risk and return required.
- Financial leverage is concerned with the relationship between a firm's EBIT and its EPS *more debt = more volatility in these*
- Thus, the use of gearing can be favorable or unfavorable
- See an example in the next slide

The Optimal Capital Structure

- Favorable effect of Leverage

An Example

- Assume that Company A and Company B are identical in every respect except in the way in which they are *capital structure* financed. Further, assume that Company A is financed entirely by equity while company B is financed 60% by equity and 40% by debt, bearing interest of 11% per year. The tax rate is 28%.

The Optimal Capital Structure

➤ Favorable effect of Leverage

An Example continued

(R millions)	Co. A	Co. B
Equity	1,000	600
Debt	-	400
Total Assets	<u>1,000</u>	<u>1,000</u>
EBIT	400.0	400.0
Interest	-	-44.0
	<u>400.0</u>	<u>356.0</u>
Tax	-112.0	-99.7
Net income	<u>288.0</u>	<u>256.3</u>
$NOPAT = EBIT(1-t)$		
Return on Assets	28.8%	28.8%
Return on Equity	28.8%	42.7%

The Optimal Capital Structure

- Favorable effect of Leverage

An Example continued

- Based on the previous slide you can see that both companies earned the same ROA which we calculated as follows: NOPAT/Total Assets
- $\text{NOPAT} = \overset{\text{op. profit}}{\text{EBIT}} \underset{\text{after tax}}{(1-t)}$
- For A --- $\text{ROA} = \{[400 * (1-0,28)]/1000\} * 100 = 28.8\%$
- For B --- $\text{ROA} = \{[400 * (1-0,28)]/1000\} * 100 = 28.8\%$
- Alternatively
- $\text{ROA for A} = (288/1000) * 100 = 28.8\%$

The Optimal Capital Structure

- Favorable effect of Leverage

An Example continued

- For company B, first calculate NOPAT
- Which is Net Income plus Interest after tax
- Net income = 256.3
- Interest after tax = $44 * (1 - 0.28) = 31.7$
- Operating income = $256.3 + 31.7 = 288$
- ROA for B = $(288 / 1000) * 100 = 28.8\%$
- ROE = $(\text{Net Profit after tax} / \text{Total Equity}) * 100$
- For A --- ROE = $\{[288 / 1000] * 100 = 28.8\%$
- For B --- ROE = $\{[256.3 / 600] * 100 = 42.7\%$

Income to debt holders

The Optimal Capital Structure

➤ Favorable effect of Leverage

An Example continued

- The Return on assets is the same for both A and B
- ROE for company B > than the ROE of company A
- The after tax cost of debt is 7.92% .i.e. $[11\% * (1 - 0.28)]$ and this has funded assets which have generated an after tax return of 28.8%.
- The net effect is that shareholders have benefited from the differential in the rates, i.e. ROA & cost of debt
- Thus, Shareholders of company B have been rewarded for the increased risk they have taken.
- This is known as positive gearing, i.e. increase in returns > than cost of debt

The Optimal Capital Structure

- Favorable effect of Leverage (continued).
- **The Impact of risk**
- Leverage increases the volatility of earnings and the above example has shown how the return to shareholders was magnified as a result of leverage.
- The converse also happens when the returns on assets is lower than the interest on debt.
- Under those circumstances, the return on shareholders' funds would be less than the return on assets.
- This is known as negative gearing. Thus, in severe cases the company may not be able to meet the interest payments from the earnings, and this may put the firm in a serious predicament which might lead to bankruptcy, see the unfavorable impact of risk in the next slide

The Optimal Capital Structure

- Unfavorable effect of Leverage.
- Consider the same companies, but in a situation in which business has been difficult and profits are only R42.5m for the year for both companies.

part of test

(R millions)	Co. A	Co. B
EBIT	42.5	42.5
Interest	-	-44.0
	42.5	-1.5
Tax	-11.9	-
Net income	30.6	-1.5
Return on Assets	3.06%	3.06%
Return on Equity	3.06%	-0.25%

÷ 1000 (arrow pointing to 30.6)
÷ 600 (arrow pointing to -1.5)

The Optimal Capital Structure

- Managers at a firm will choose a mix of debt and equity in their capital structure that minimises the cost of financing the firm's projects, i.e. they will choose a mix of debt and equity that will minimise their cost of capital
- Since the optimal capital structure is a mix of debt and equity that minimises the cost of financing the firm's projects, it also means that the optimal capital structure is that mix of debt and equity that if used in the firm, will result in the maximization of the total value of the firm's projects and, hence, the overall value of the firm.

The Modigliani and Miller Approach

Modigliani and Miller Proposition 1

- This approach initially stated that there is no optimal capital structure as the increased use of cheaper debt will be netted off against the more expensive equity.
- Equity would become more expensive for any increases in debt, because shareholders would be exposed to higher risk, so demand a higher return.
- The effect is that WACC stays constant. Thus, the value of the business is dependent on the assets and not the way in which the business is financed.
- The value of the business = $\text{EBIT} / \text{WACC}$ *-perpetuity calc.*
- M-M proved their theory by establishing that if the WACC was not constant then arbitrage would take place and drive the prices into equilibrium.

The Modigliani and Miller Approach

Modigliani and Miller Proposition 1

➤ M-M made certain assumptions in reaching their initial conclusions: *perfect world, perfect info, no gov, no taxes*

- Individuals can borrow and lend at the same rates as those awarded to companies
- Perfectly efficient markets exists.
- No transaction and information costs
- There are no income taxes.
- No costs associated with financial distress
- There are no agency costs
- Debt is free and cost of debt remains constant for all levels of debt
- Market participants have similar expectations and agree on future earnings of companies.

The Modigliani and Miller Approach

Modigliani and Miller Proposition 1

➤ **Net Income Approach** *Not asking this one*

➤ Assumes that financial leverage has no impact on the K_e or K_d .

➤ Thus, the greater the debt, the lower the WACC, and the higher the value.

➤ **Net Operating Income Approach (MM)**

➤ Assumes that K_e increases with financial leverage but that the K_d and WACC remain constant.

➤ Thus financial leverage has **no impact on value**.

The Modigliani and Miller Propositions

- The real investment policy of the firm includes the criteria that the firm uses in deciding which real assets (projects) to invest in.
- The market value of the debt plus the market value of the equity must equal the value of the cash flows produced by the firm's assets.

$$V_{\text{Firm}} = V_{\text{Assets}} = \overset{\text{how it's financed}}{V_{\text{Debt}}} + V_{\text{equity}} \quad (16.1)$$

The Modigliani and Miller Propositions

- Modigliani and Miller Proposition 1 essentially says that the combined value of the equity and debt claims (represented by the present value of free cash flows the firm's assets are expected to produce in the future) does not change when you change the capital structure of the firm if no one other than the shareholders and the debt holders is receiving cash flows.

The Modigliani and Miller Propositions

- Such a change is called a financial restructuring, where a combination of financial transactions occur that change the capital structure of the firm without affecting its real assets.

$$\begin{array}{r} E \\ D \end{array} \begin{array}{r} 1000 \\ 0 \end{array} \quad \begin{array}{r} 600 \\ 400 \end{array} \quad \left. \vphantom{\begin{array}{r} E \\ D \end{array}} \right\} \Delta \text{ shouldn't } \Delta \text{ value of firm}$$

Exh I 6.1: Capital Structure and Firm Value under M&M Proposition 1

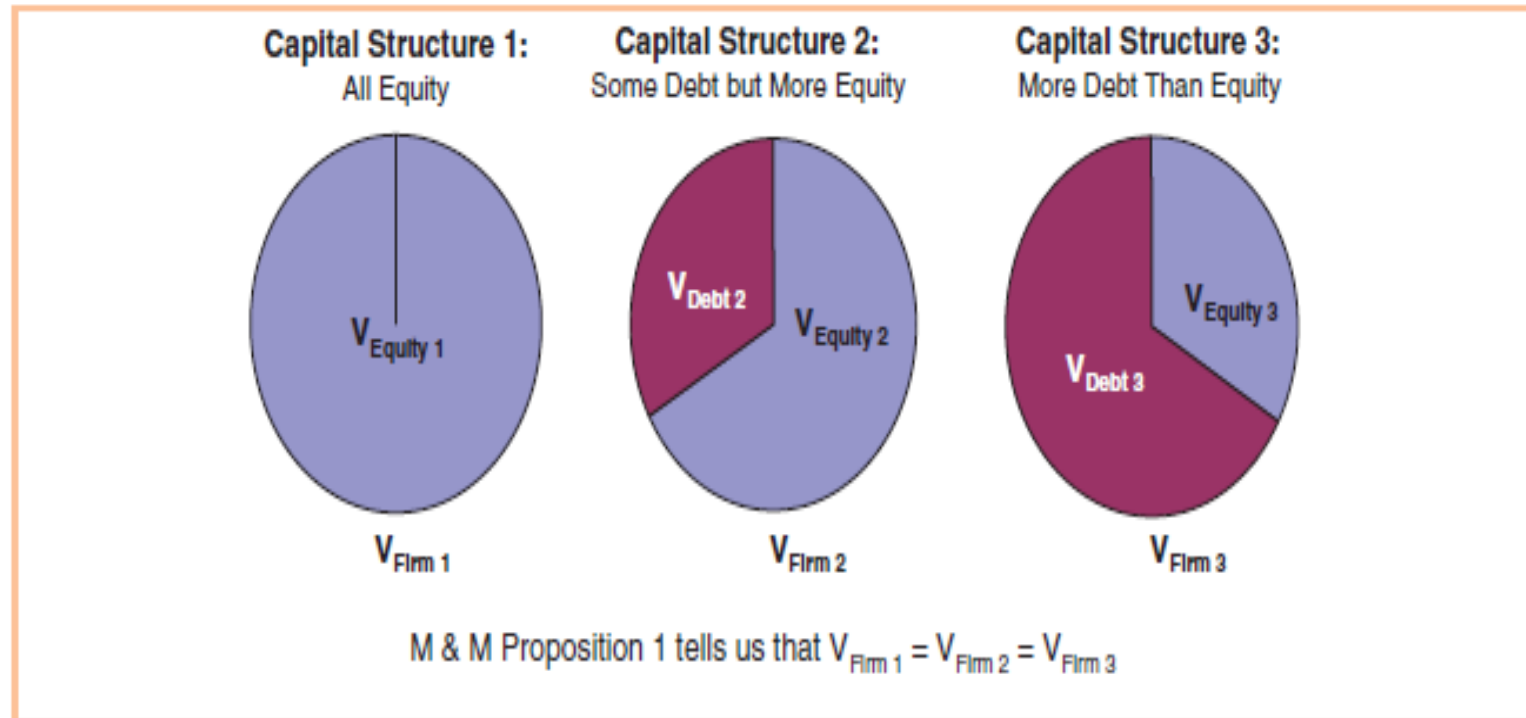


Exhibit 16.1: Capital Structure and Firm Value under M&M Proposition 1 The size of the pie represents the present value of the cash flows that the assets of a firm are expected to produce in the future (V_{Firm}). The sizes of the slices reflect the value of the total cash flows that the debt holders (V_{Debt}) or shareholders (V_{Equity}) are entitled to receive for three different capital structures. Under the three conditions identified by M&M, the total value of the cash flows to the firm does not change, regardless of which capital structure the firm uses.